

TECHNICAL PRESENTATION

# DIAL-ALERT

Early prediction of blood pressure defined  
intradialytic hypotension

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Capstone Project, Step 6  
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Academic clinical decision support prototype

# Prediction question and evaluation measures

A supervised binary-classification task with a fixed prediction boundary

**Can information available near the start of haemodialysis identify sessions that later record systolic blood pressure below 90 mmHg?**

## POPULATION

Eligible adult  
haemodialysis sessions

## PREDICTION TIME

Earliest valid active  
record within minutes 0  
to 30

## TARGET

Any later SBP below 90  
mmHg

## DECISION USE

Risk ranking for clinician  
review

## Evaluation hierarchy

Primary ranking

**Average precision**

Discrimination

**ROC AUC**

Probability quality

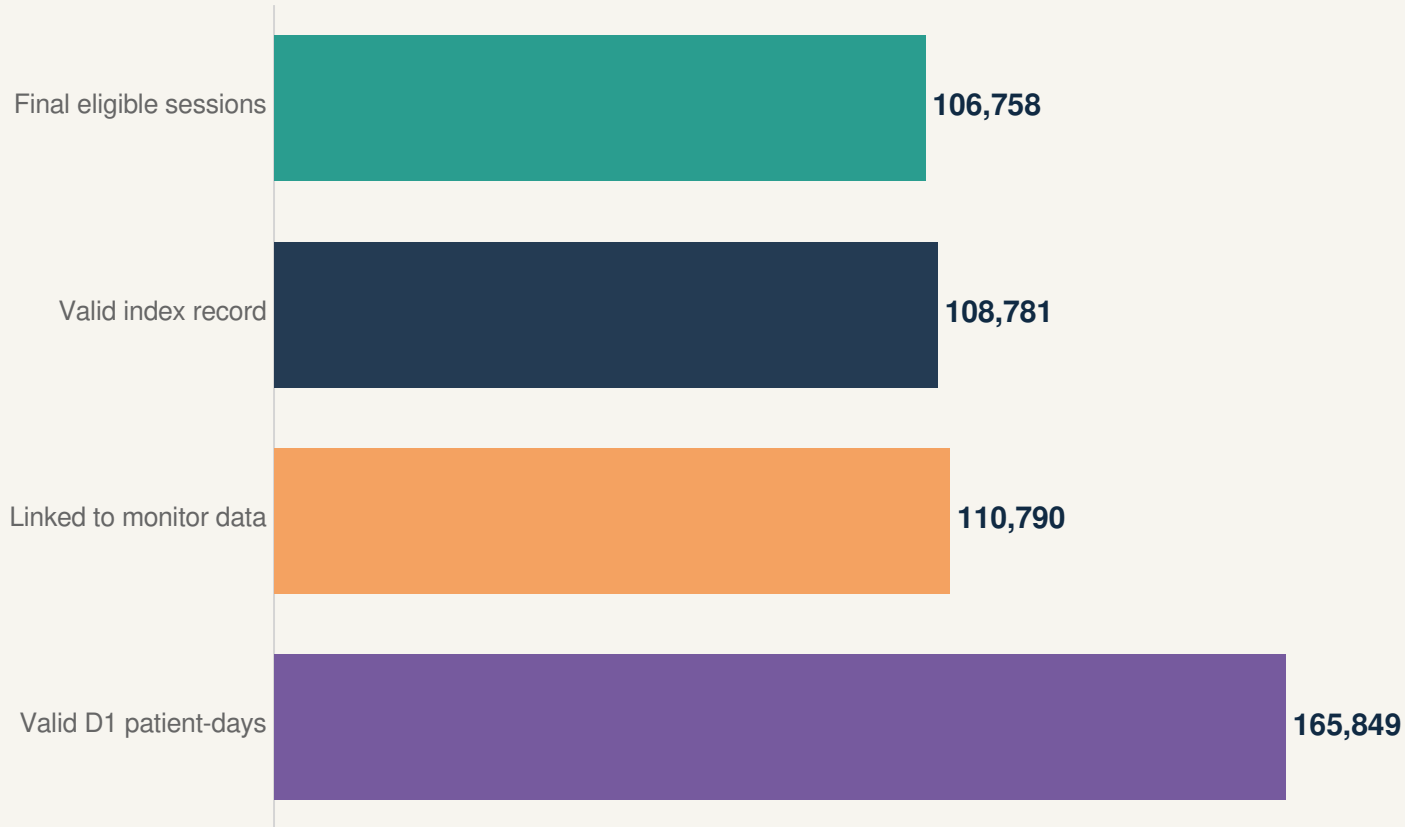
**Brier score**

Workflow utility

**Recall and PPV at alert  
capacity**

# HEMOBP cohort and eligibility

Public longitudinal data link patient, session, and monitor records



**106,758**

eligible sessions

**830**

unique patients

**8.49%**

primary outcome prevalence

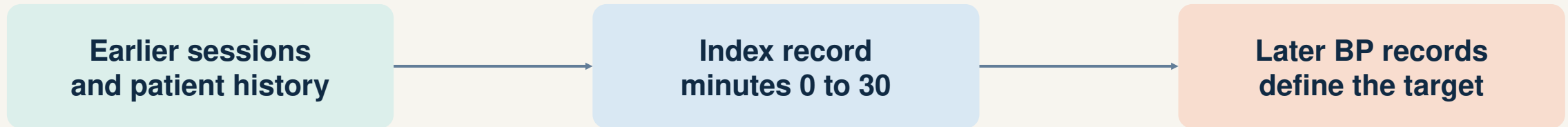
**4.37 million monitor states in the raw source**

Eligibility: baseline SBP  $\geq 90$ , two later measurement minutes, last dialysis minute  $\geq 120$

*Minute 120 is elapsed dialysis time, not 120 minutes after the index reading*

# Prediction features stop at the index time

The outcome uses later measurements only, preventing temporal leakage



## 22 model inputs

### Haemodynamic state

Baseline SBP, DBP, MAP, pulse pressure

### Fluid and treatment

Fluid excess, UF rate, blood flow, dialysate settings

### Longitudinal history

Prior nadir SBP, prior IDH rate, prior session event

### Patient context

Age, recorded sex, diabetes, dialysis vintage

**Excluded from predictors: later blood pressure, outcome fields, post-dialysis weight, dialysis end time, identifiers, and audit variables**

Source: preprocessing specification and feature engineering dictionary

# Patient-level separation protects the evaluation

No patient appears in more than one development partition

496

training patients  
64,053 sessions

164

validation patients  
21,351 sessions

170

test patients  
21,354 sessions

0 patient overlap

1

## Grouped 3-fold CV

Tune candidate pipelines on  
training patients

2

## Validation decisions

Choose calibration and F2  
threshold

3

## Locked test

Evaluate once after all  
choices

4

## Patient bootstrap

500 resamples for  
uncertainty

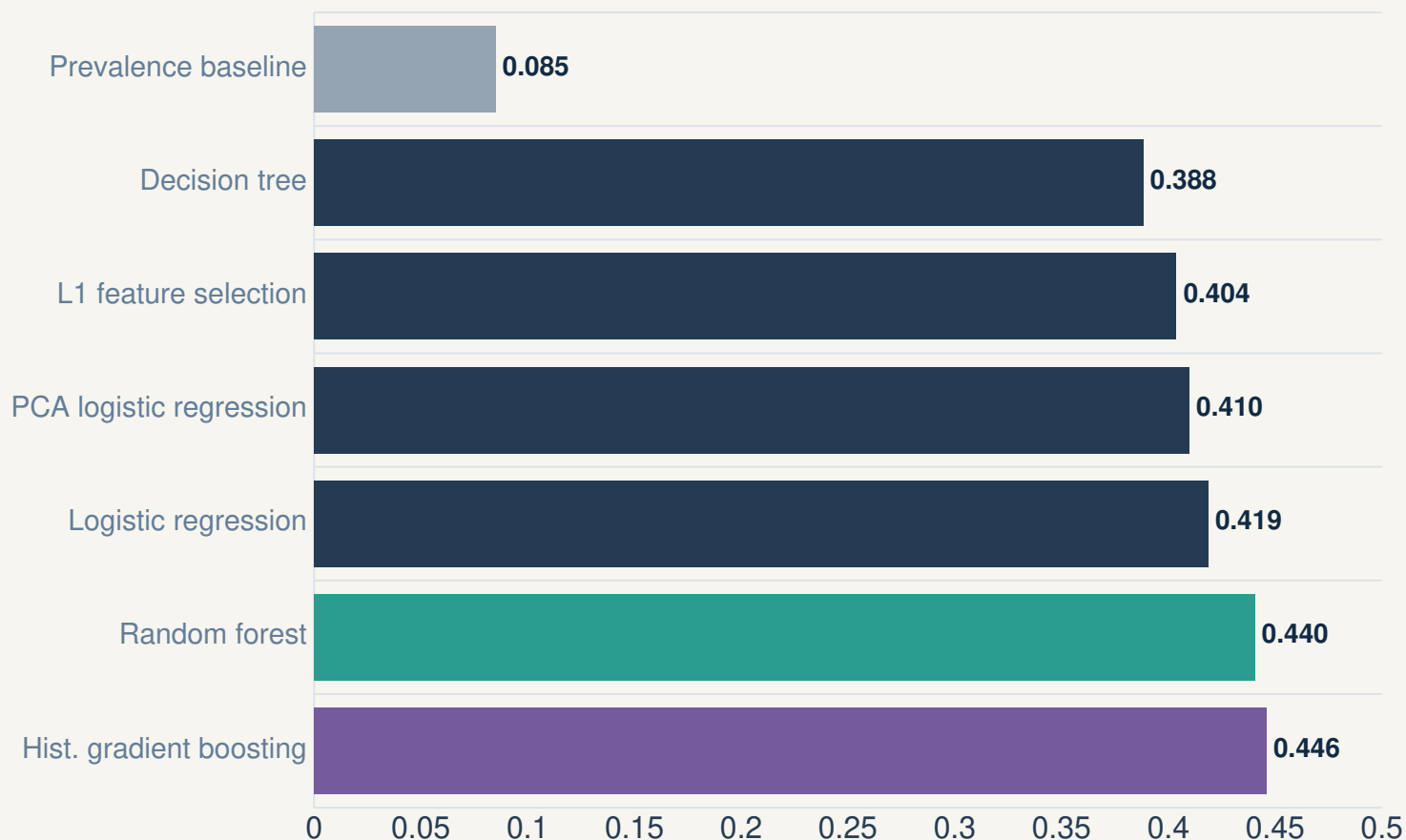
## Why it matters

Repeated sessions from the same patient can make random session splits look more accurate than true generalisation to new patients.

Source: [split\\_summary.csv](#), [split\\_assignments.csv.gz](#), [model\\_config.json](#)

# Why random forest was selected

Selection rule: highest grouped-CV average precision, then lower validation Brier score within 0.01



PCA and prevalence baseline are comparison benchmarks outside the final selection pool

## Why random forest

CV AVERAGE PRECISION

**0.440**

VALIDATION BRIER SCORE

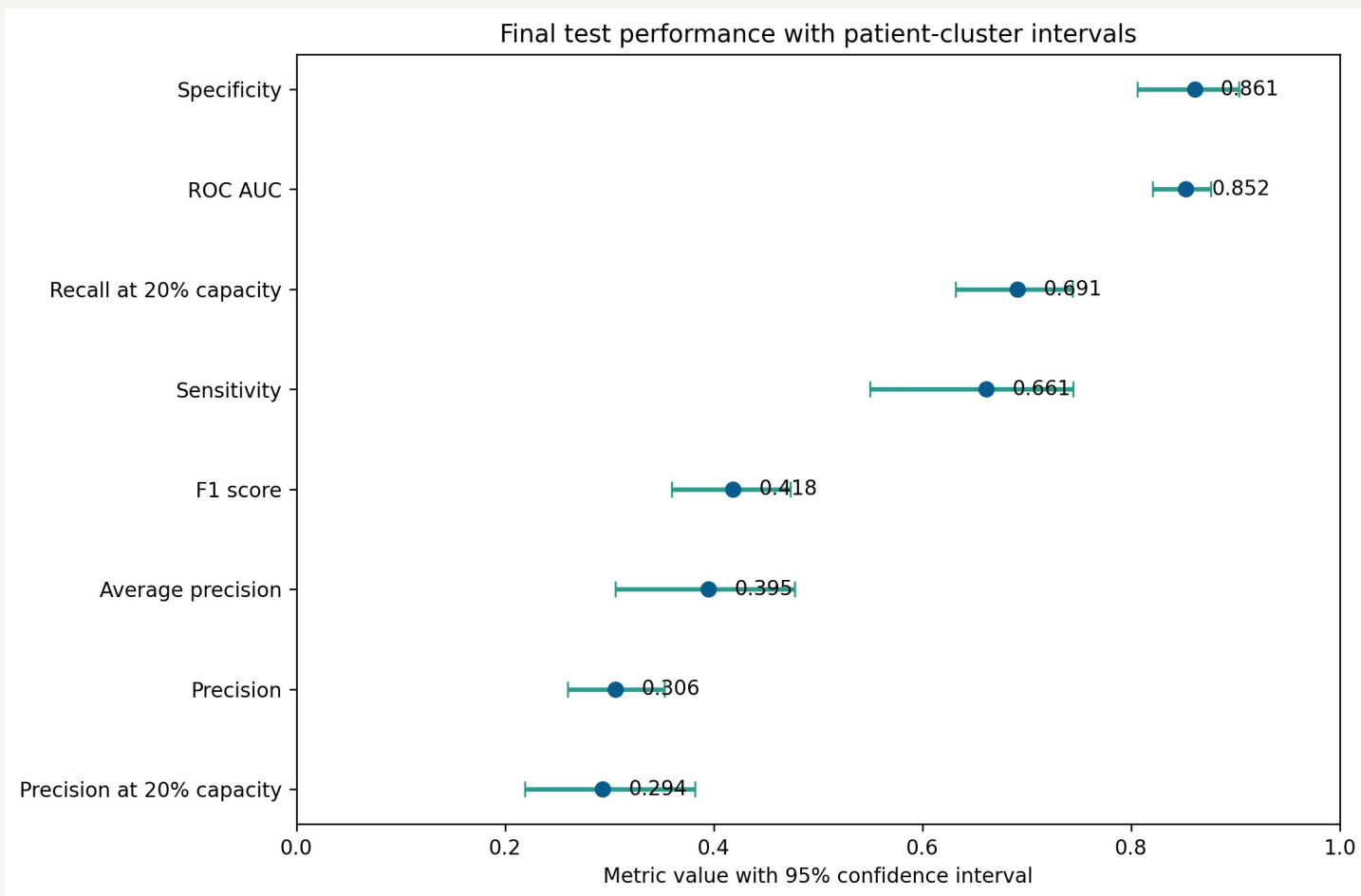
**0.058**

Histogram boosting ranked slightly higher on CV AP (0.446), but its validation Brier score was 0.134. The difference in AP was within the predefined 0.01 tolerance.

**Selected model: 250-tree random forest, minimum leaf size 15**

# Test performance and uncertainty

Locked assessment on 21,354 sessions from 170 previously unseen patients



**0.852**

ROC AUC

**0.395**

average precision

**0.062**

Brier score

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**Operating threshold 0.143**

Sensitivity 66.1%

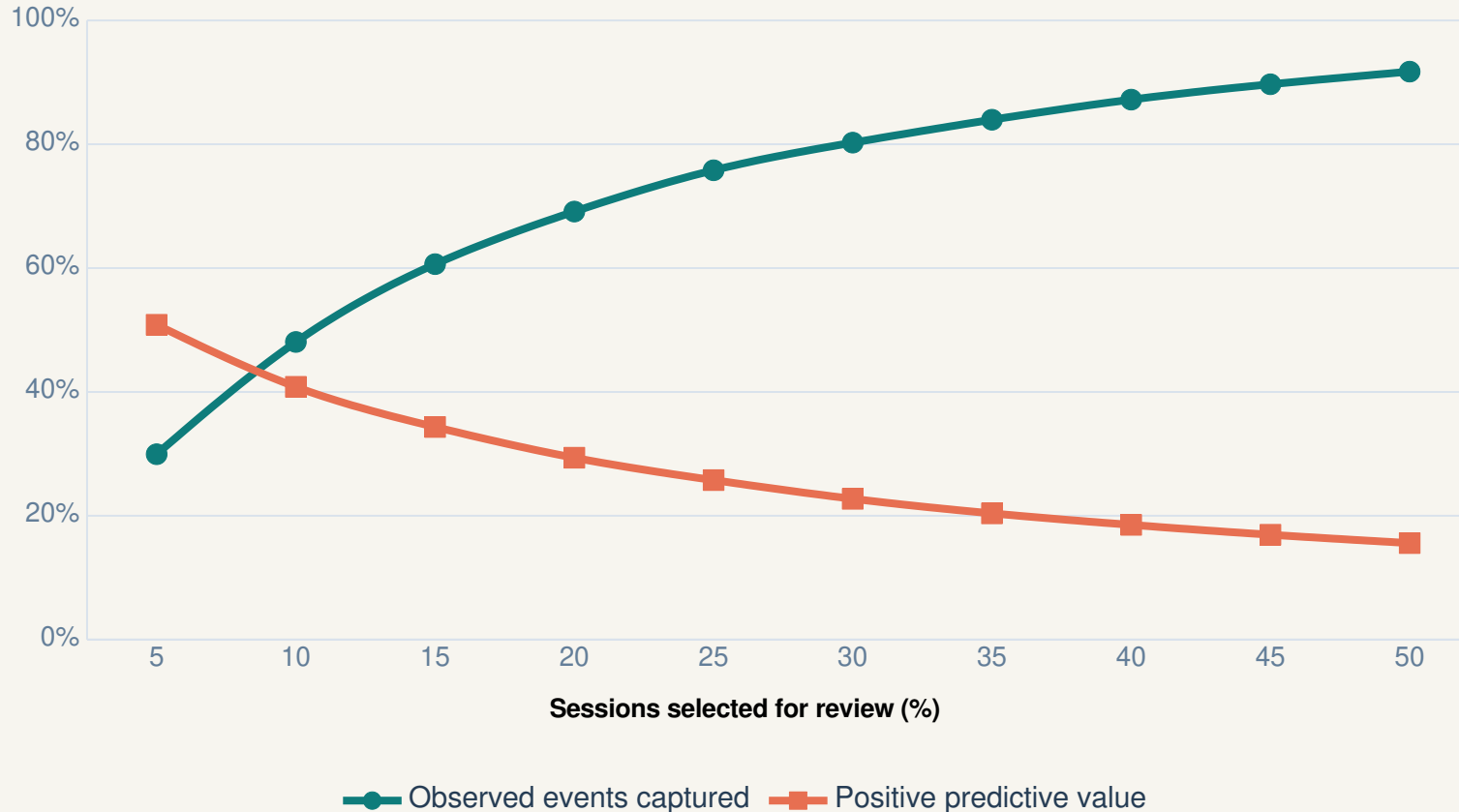
Specificity 86.1%

Precision 30.6%

*Intervals: 500 patient-cluster bootstrap resamples*

# Review capacity and event detection

Ranking utility shown across the proportion of sessions selected for review



**20% review capacity**

EVENT RECALL

**69.1%**

POSITIVE PREDICTIVE VALUE

**29.4%**

LIFT OVER PREVALENCE

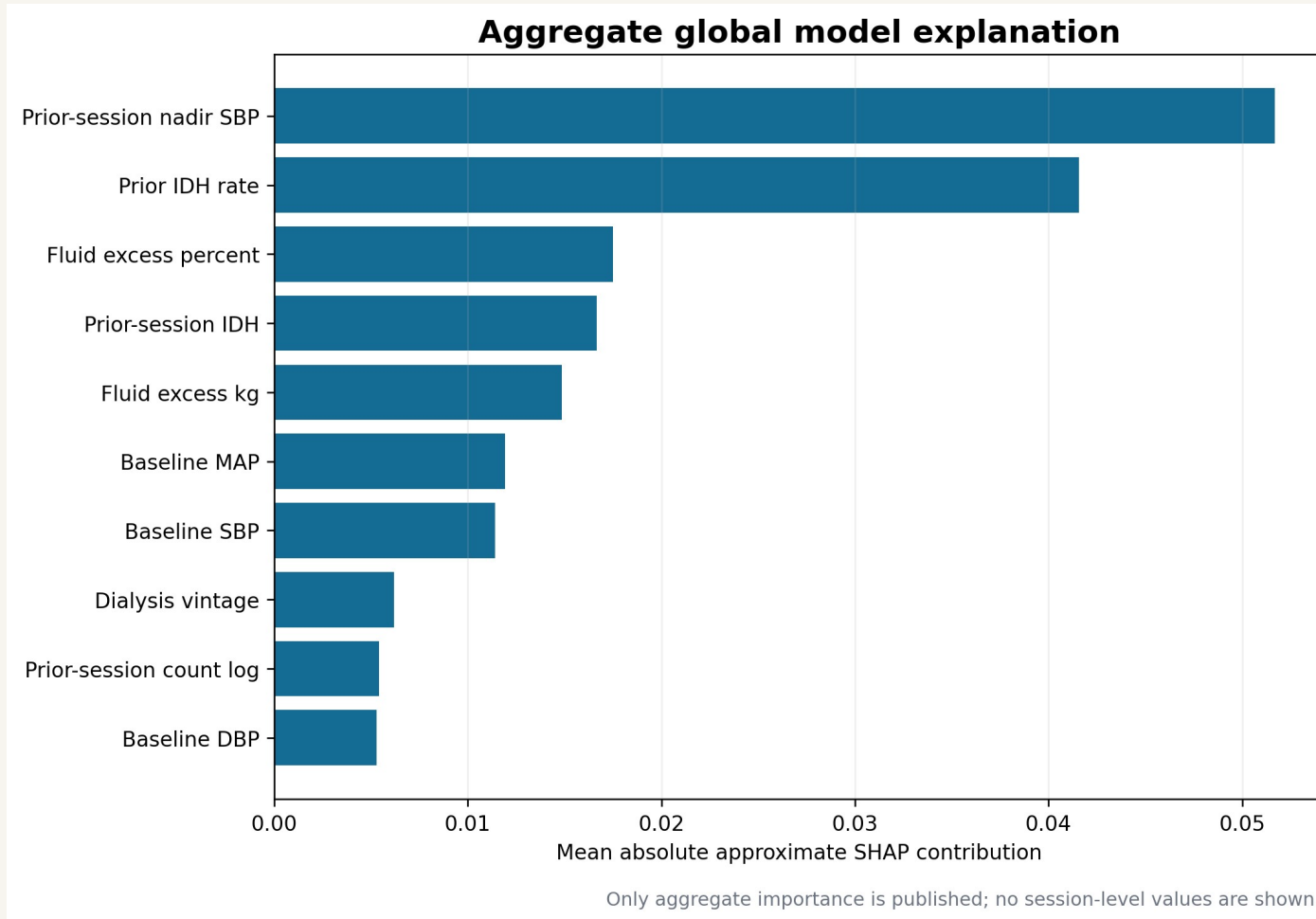
**3.46×**

**14.1 false alerts per 100 sessions**

Separate fixed threshold 0.143: 18.35% flagged, 66.1% sensitivity, 30.6% precision

# Model explanations and their limits

Aggregate SHAP with a separately declared synthetic local scenario



## Consistent signals

**Lower prior nadir SBP:** higher predicted risk

**Higher prior IDH rate:** higher predicted risk

**Greater fluid excess:** higher predicted risk

**Lower baseline pressure:** higher predicted risk

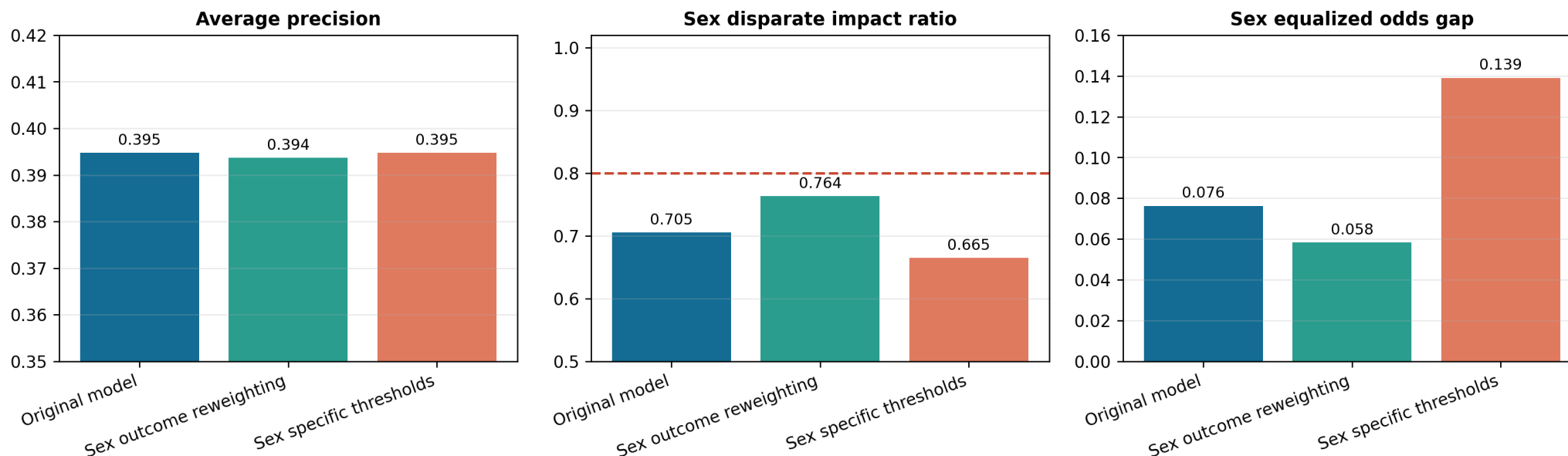
## Explanation limits

Monte Carlo SHAP is approximate. For the synthetic local scenario, the LIME-style surrogate had weighted  $R^2$  0.678 and absolute probability error 0.025. Neither method establishes causality or a treatment target.

# Fairness mitigation results

Held-out fairness metrics must be reviewed with clinical utility and uncertainty

## Mitigation changes fairness and performance together



### Reweighting retained

Sex disparate-impact ratio 0.705 to 0.764.  
Equalized-odds gap 0.076 to 0.058. AP  
0.395 to 0.394.

### Threshold strategy rejected

Sex-specific thresholds worsened the  
equalized-odds gap to 0.139 and lowered  
precision to 27.8%.

### Unmeasured fairness

Race, ethnicity, socioeconomic status, and  
gender identity are unavailable. Age-group  
DI ratio is 0.541.

*0.80 disparate-impact ratio is a descriptive screen, not a fairness certificate*

# Reproduction instructions and verification status

Saved-model inference verified; full training reproduction blocked at source download

## Project structure

|                             |                                                   |
|-----------------------------|---------------------------------------------------|
| <code>data/raw</code>       | Download locally; excluded from Git               |
| <code>data/processed</code> | Leakage-controlled session table                  |
| <code>src</code>            | Build, train, evaluate, audit, and report scripts |
| <code>configs</code>        | Seeds, features, CV, bootstrap, capacity          |
| <code>models</code>         | Selected predictor; candidates regenerate         |
| <code>artifacts</code>      | Aggregate metrics; assignments regenerate         |
| <code>reports</code>        | Step 2 through Step 6 deliverables                |

## Reproduction sequence

- 1 `python src/build_session_dataset.py`
- 2 `python src/train_evaluate.py`
- 3 `python src/generate_eda.py`
- 4 `python src/audit_bias_fairness.py`

**Model manifest records package versions, file hashes, random seeds, feature order, and threshold metadata.**

# Internal validation supports a governed next study

The evidence is promising for risk ranking and insufficient for clinical deployment

## Evidence supports

**Risk ranking:** ROC AUC 0.852 and AP 0.395

**Capacity planning:** 69.1% event recall at 20% review

**Interpretability:** clinically coherent history and fluid signals

**Mitigation candidate:** sex-outcome reweighting

## Evidence does not support

- Clinical benefit or cost savings

- Autonomous diagnosis or treatment

- Fairness across unavailable attributes

- Transfer to another centre or future protocol

## Next validation gates

External temporal and geographic validation, prospective silent mode, human factors, subgroup calibration, monitoring, and rollback